

Safety and Ethics: Building Trustworthy Robot Experiences

A fictional course sample about practical safeguards and thoughtful choices.



Figure 1. Original watercolor-style illustration created for this fictional course sample.

Why this topic matters

A robot project is successful only when people can use it safely and understand what it is allowed to do. Safety concerns physical movement, but it also includes privacy, clear communication, and a way to report problems. Ethical questions make those design choices visible instead of treating them as an afterthought.

A short fictional scene

In a fictional school workshop, a robot pauses at a marked boundary before approaching a shared table. A facilitator can stop the activity with one button. The team also explains which sensor data is stored and which is discarded. These small choices show respect for the people around the machine.

How the idea works

Good questions are concrete. Who can activate the robot? What happens if a sensor disagrees? Can someone opt out of being recorded? How will the team notice an unfair or confusing result? The answers should be tested in the same calm way as the robot's motion.

A practical way to observe it

In a fictional school workshop, a robot pauses at a marked boundary before approaching a shared table. A facilitator can stop the activity with one button. The team also explains which sensor data is stored and which is discarded. These small choices show respect for the people around the machine. In a classroom demonstration, learners can pause after each step and describe what information the robot needs before it continues. This turns a smooth-looking robot action into a sequence of visible decisions.

Fictional trust checklist

Area		
Physical safety	Can motion stop quickly?	Accessible stop control
Privacy	Is data necessary?	Collect only task data
Transparency	Can people understand it?	Visible status and plain language

Questions for a parsing exercise

Can a document parser recognize the section headers, the rows and columns of the table, the page footer, and the figure caption? Compare the extracted structure with the visual layout before deciding whether an element was parsed correctly.

Invented learning data

The chart below uses deliberately fictional values. Its purpose is to give a document parser a distinct visual data element, not to represent a real benchmark or product claim.

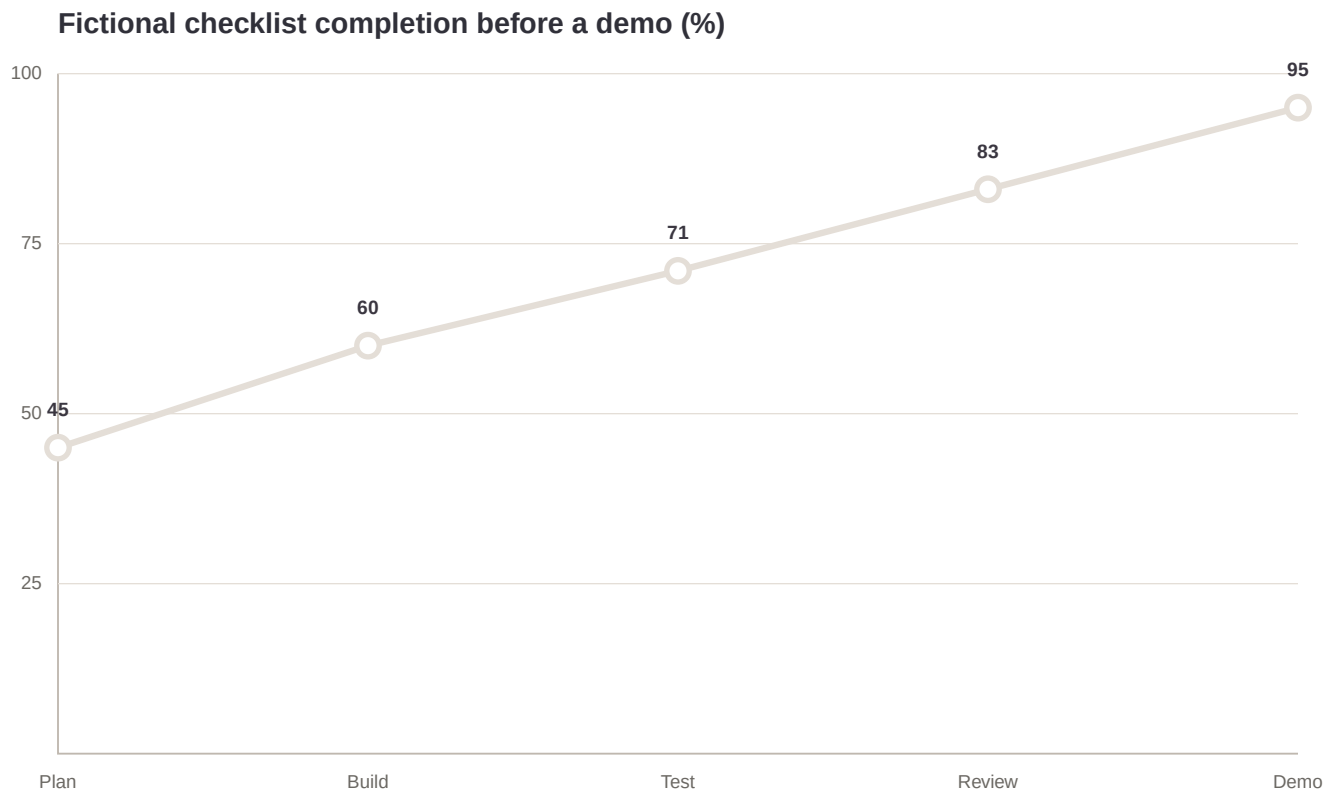


Figure 2. Fictional data for visual parsing practice; values are not measured results.

Key terms for discussion

A structured look at possible harms and controls.

Making a system's purpose and limits understandable.

A person's informed agreement to take part.